# Image Processing and Medical Imaging

## Medical Imaging

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# Medical Imaging

- Medical imaging is **non-invasive visualization** of internal organs, tissue, etc.
- Image can be 2D or 3D. It can be still image or video.
- Endoscopy and microscopy are not imaging modality.

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# Major Modalities

- Projection X-ray (Radiography)
- X-ray Computed Tomography (CT)
- Nuclear Medicine (SPECT, PET)
- Ultrasound
- Magnetic Resonance Imaging

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# Projection X-ray Imaging

- X-ray imaging requires interactions of x-ray photons with object – work in a specific energy band.
  - Above this band – body is too transparent
  - Below this band – body is too opaque
  - Well below this band – wavelengths are too long
- One problem with x-ray: no depth information.
- The result is a 2D image with good resolution.

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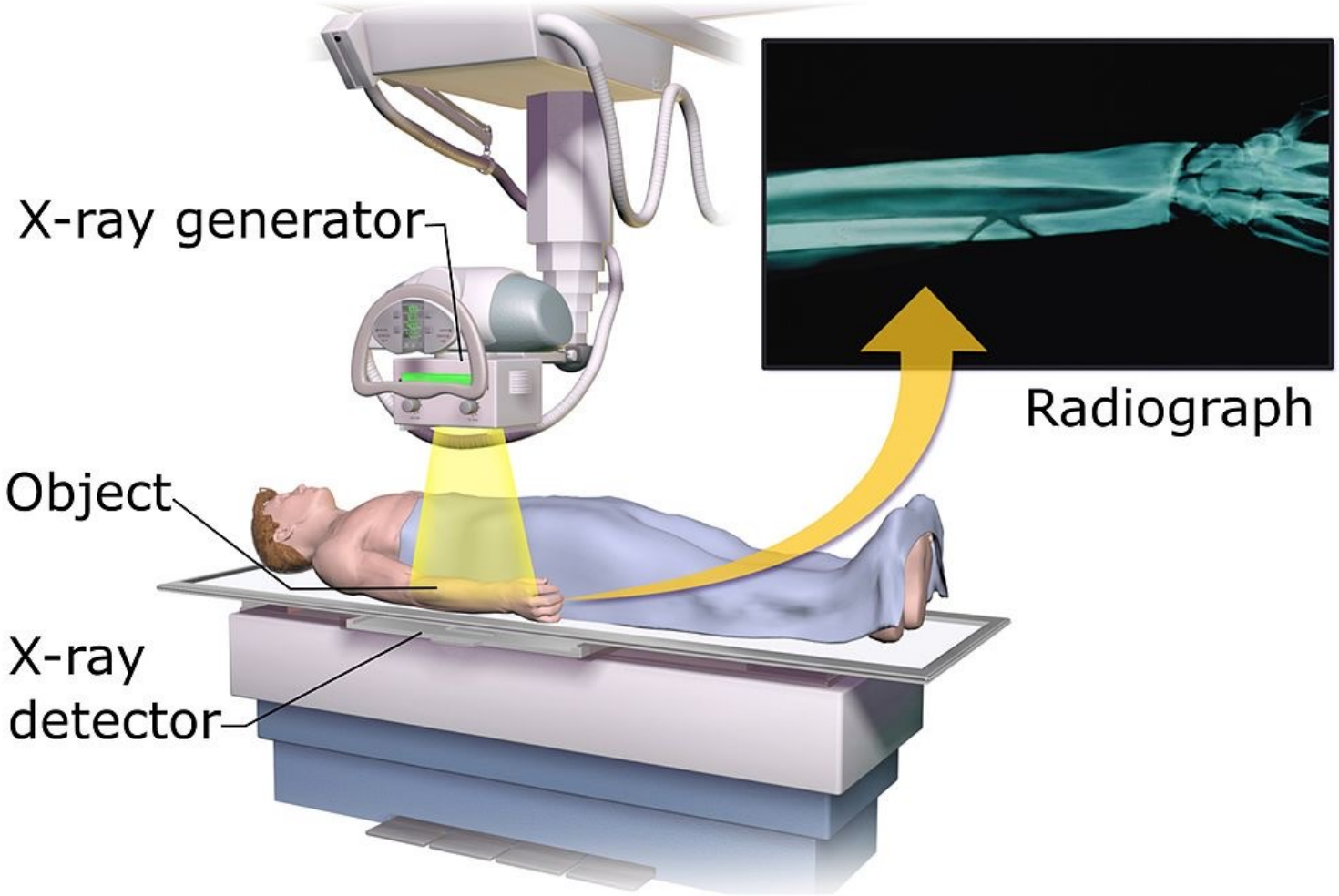
# Projectional radiography

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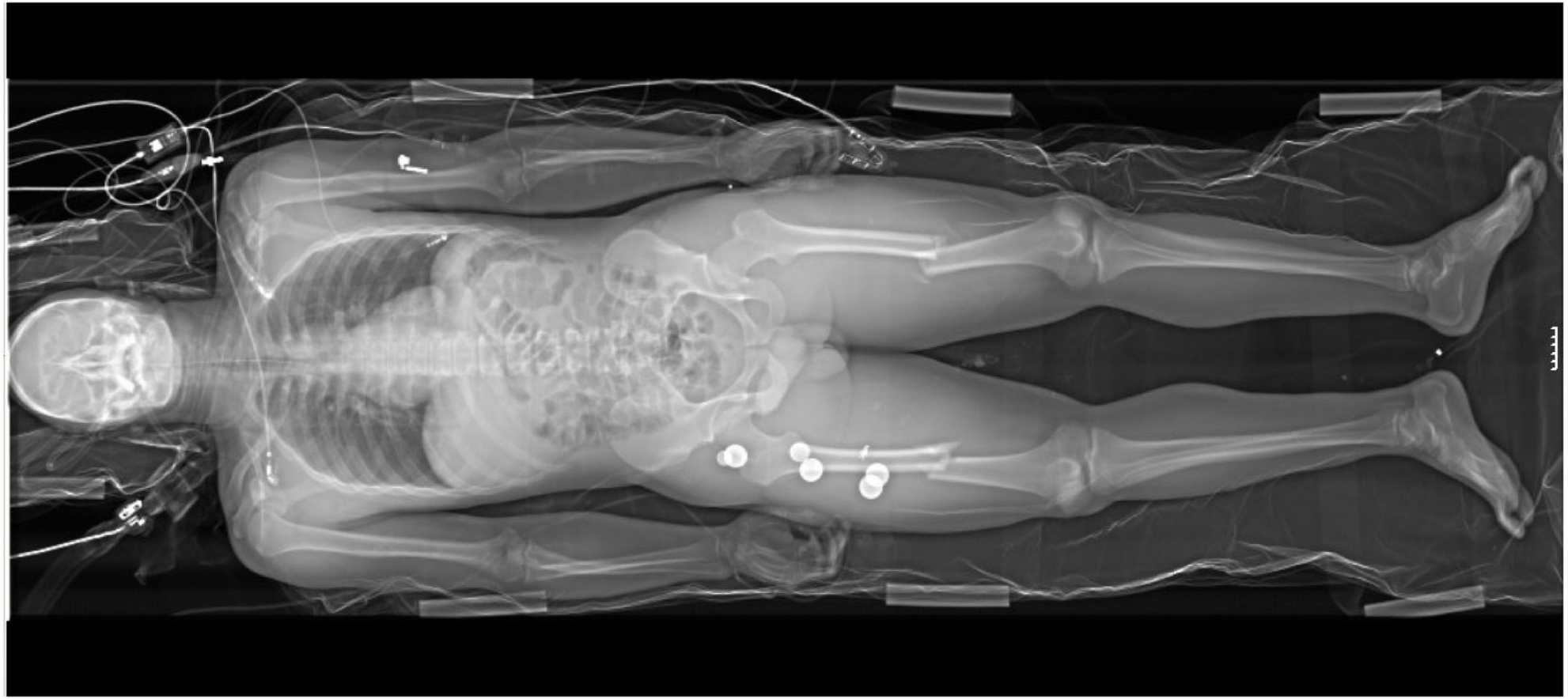
Source: By Blausen MedicalAnnotations by Mikael Häggström - By Blausen Medical., CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=58054654

# Projection X-ray Imaging



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Source: Dimitrios S Evangelopoulos, Simone Deyle, Heinz Zimmermann and Aristomenis K Exadaktylos / CC BY (https://creativecommons.org/licenses/by/2.0)

# X-ray Computed Tomography (CT)

- Uses X-rays, but exposure is limited to a slice (or “a couple of” slices) by a collimator
- Source and detector rotate around object – projections from many angles
- The desired image, $I(x,y) = \mu(x,y,z_0)$, is computed from the projections
- There is depth information.

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# X-ray Computed Tomography (CT)

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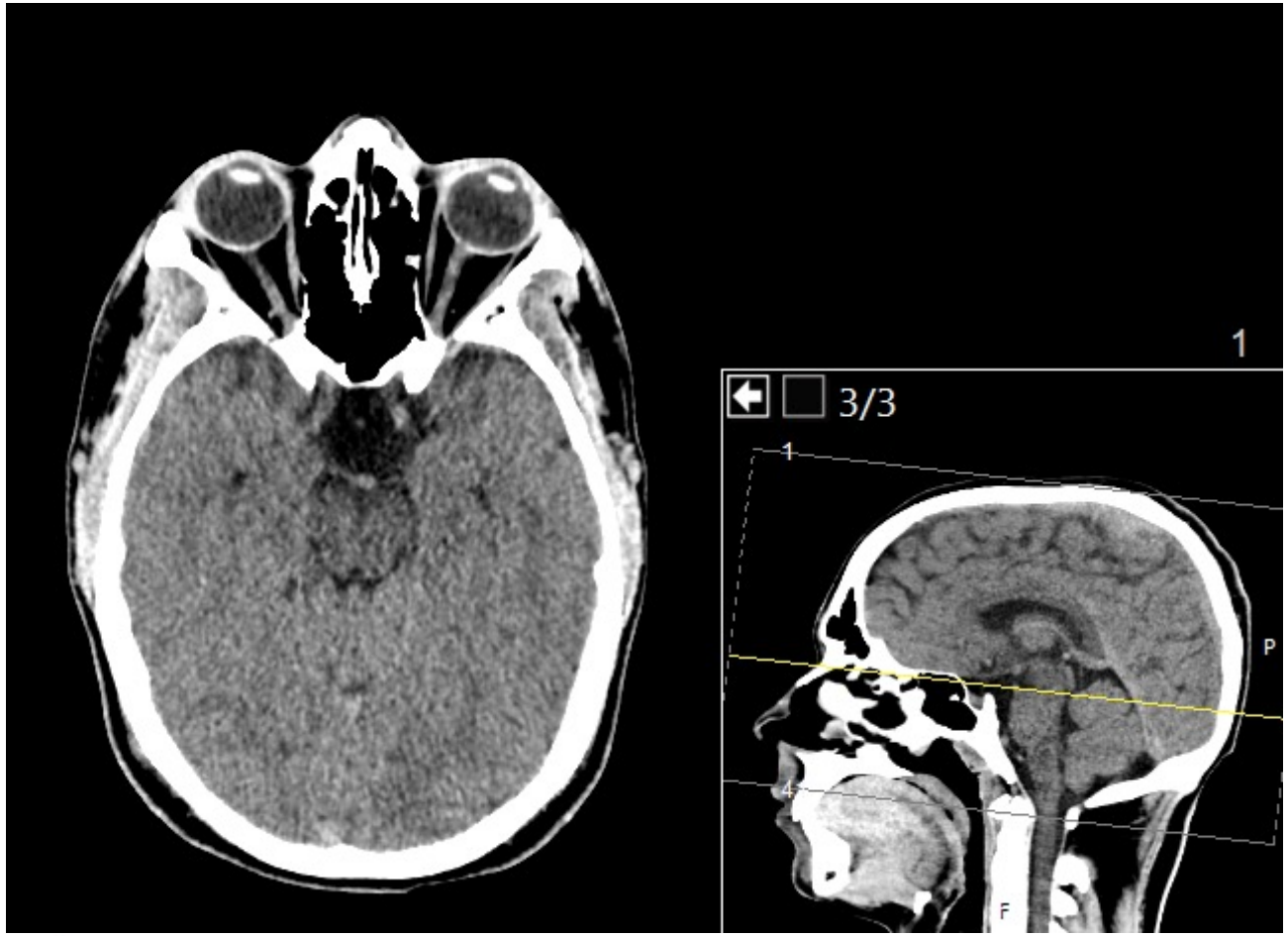
Source: MBq, Copyrighted free use, https://commons.wikimedia.org/w/index.php?curid=32847845

# X-ray Computed Tomography (CT)

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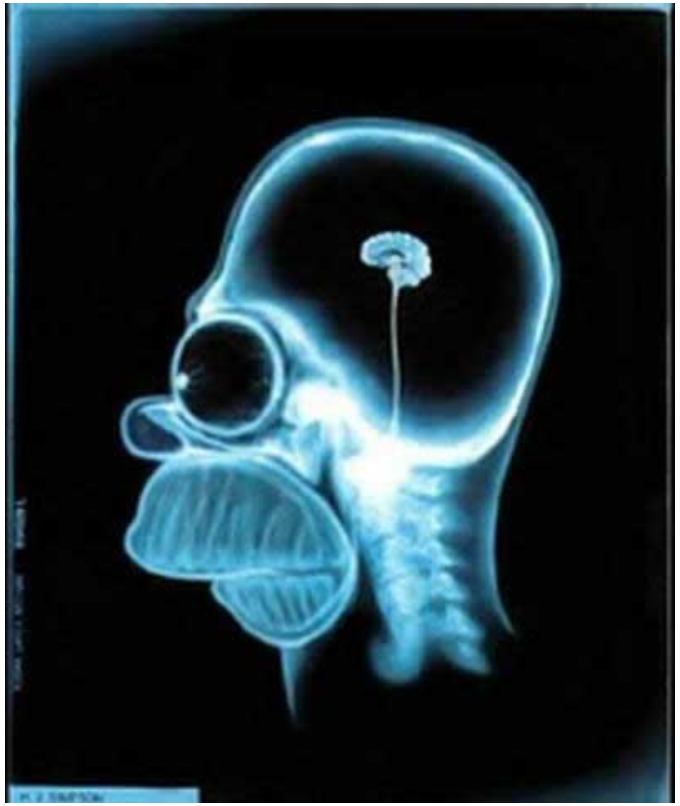
Source: https://commons.wikimedia.org/wiki/Scrollable_computed_tomography_images_of_a_normal_brain_(case_1)

# Anatomical Imaging vs. Functional Imaging



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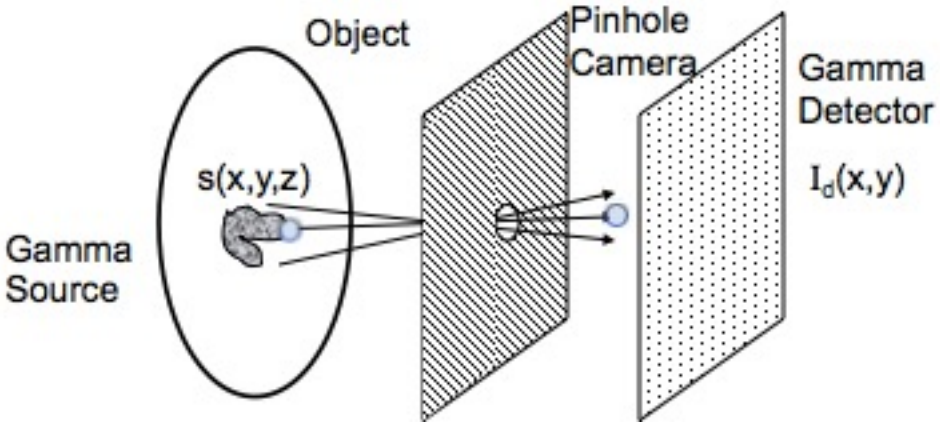
Source: https://www.slideshare.net/khalednagib/el-nefiawy-nagwa-radiographic-anatomy

# Nuclear Medicine (Scintigraphy)




- Detector records *emission* of gamma photons from radioisotopes introduced into the body.
- The integral is a line-integral or a "projection" through object.
- Source s(x,y,z) usually represents a selective uptake of a radio-labeled pharmaceutical.

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# Nuclear Medicine (cont.)

- Issue: Pinhole Size
  - Large pinhole – more photons, better SNR
  - Large pinhole – more blur, reduced resolution
- Issue: Half-life
  - Long half lives are easier to handle, but continue to irradiate patient after imaging is done.
- Issue: Functional Specificity
  - Pharmaceuticals must be specific to function of interest
  - E.g. Thallium, Technicium
- Issue: No depth info
  - Nuclear Medicine Computed Tomography (SPECT, PET)

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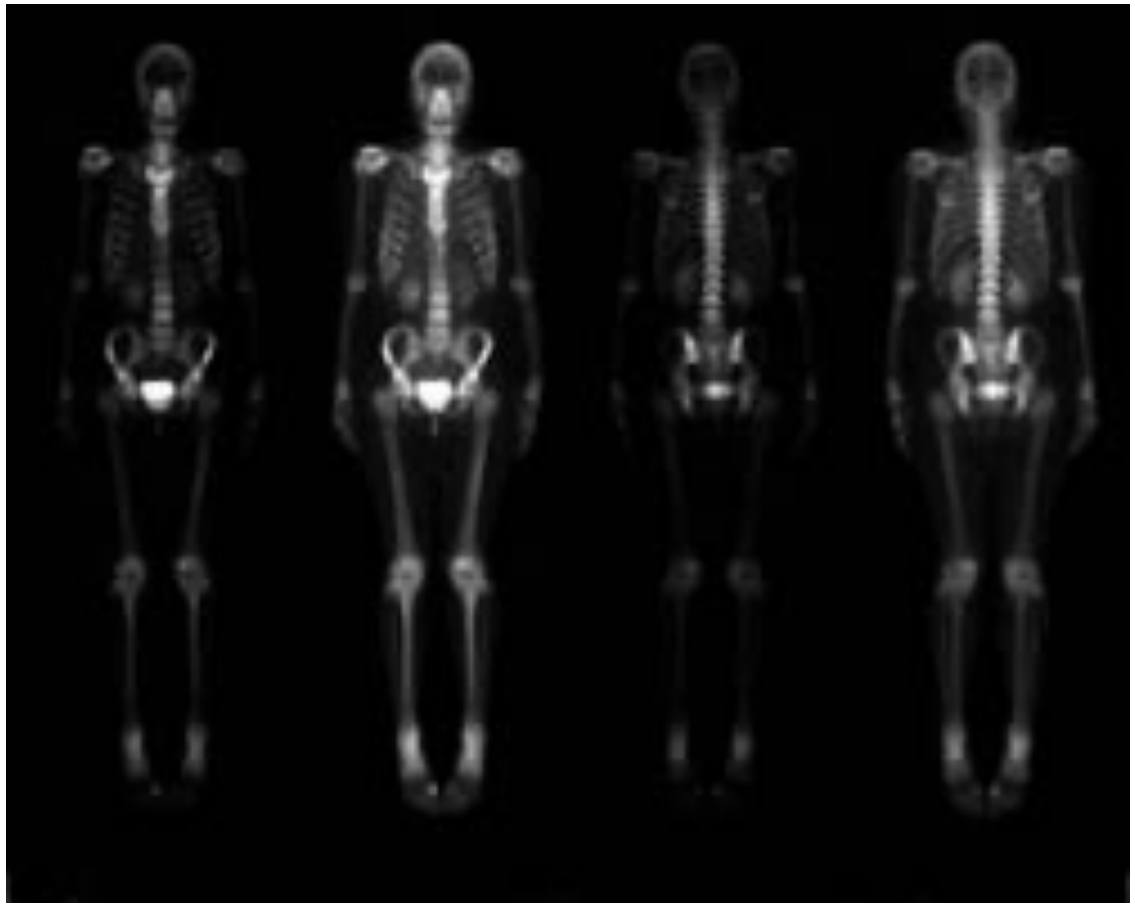
# Nuclear Medicine (cont.)



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# Positron-Emission Tomography (PET)

- **Positron-emission tomography (PET)** is a nuclear medicine functional imaging technique that is used to observe metabolic processes in the body.
- The system detects pairs of gamma rays emitted indirectly by a positron-emitting **radioligand**, most commonly Fluorine-18, which is introduced into the body on a biologically active molecule called a **radioactive tracer**.
- Different ligands are used for different imaging purposes, depending on what the radiologist wants to detect.

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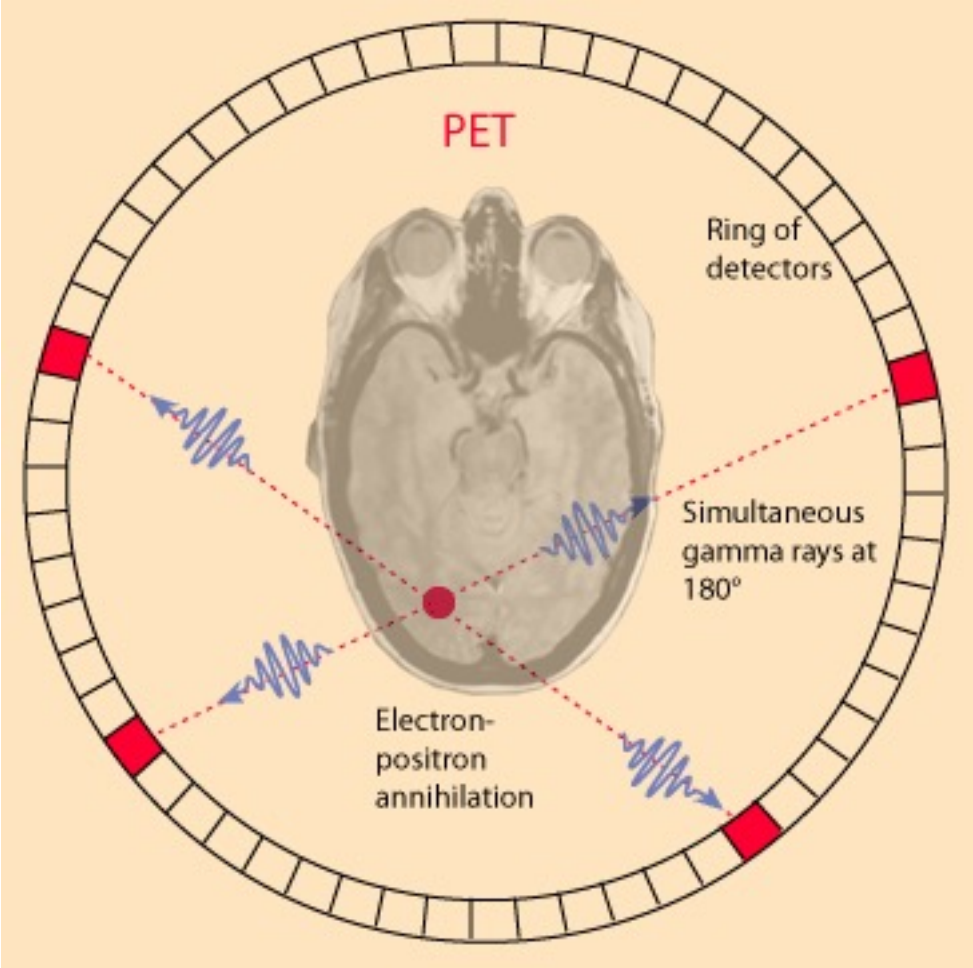
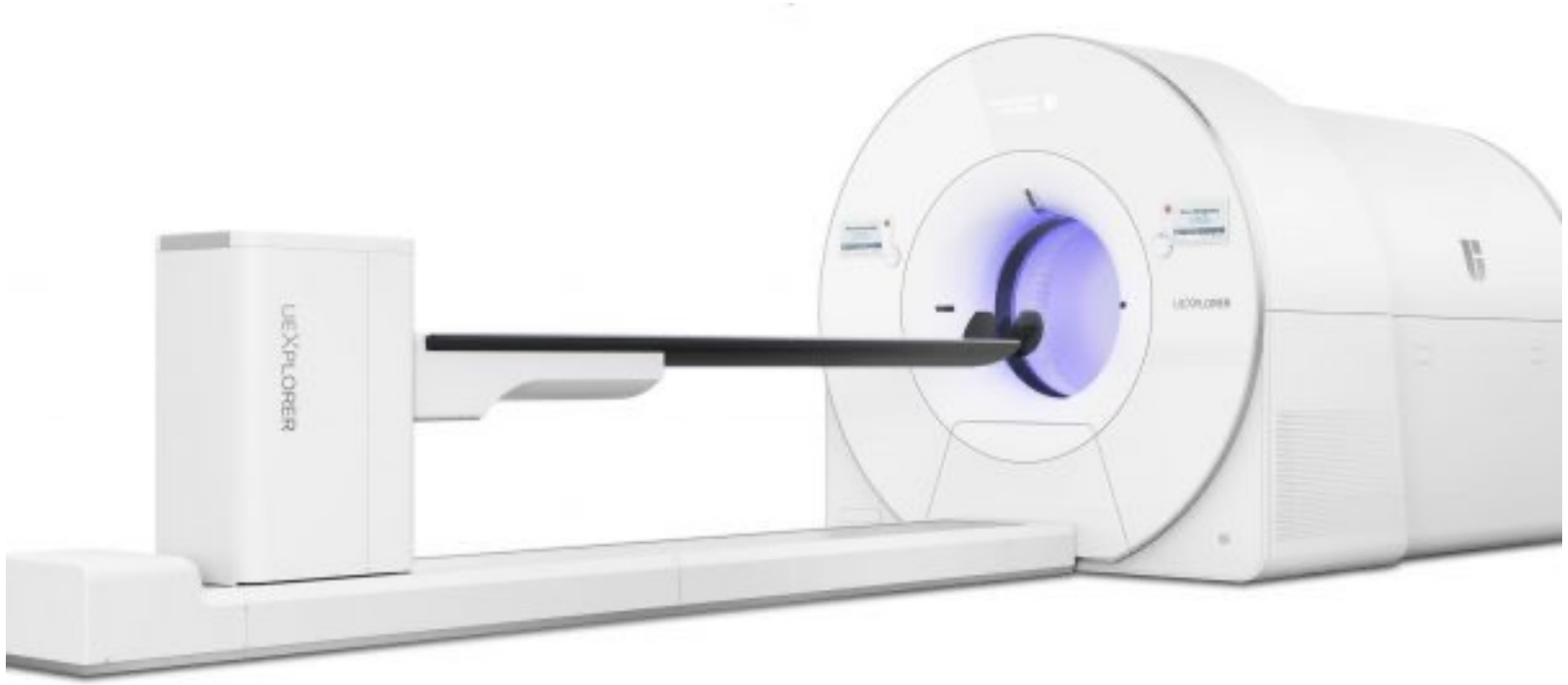
Source: Wikipedia

# PET Scanner




Source: http://hyperphysics.phy-astr.gsu.edu/hbase/NucEne/nucmed.html

# PET-CT Scanner

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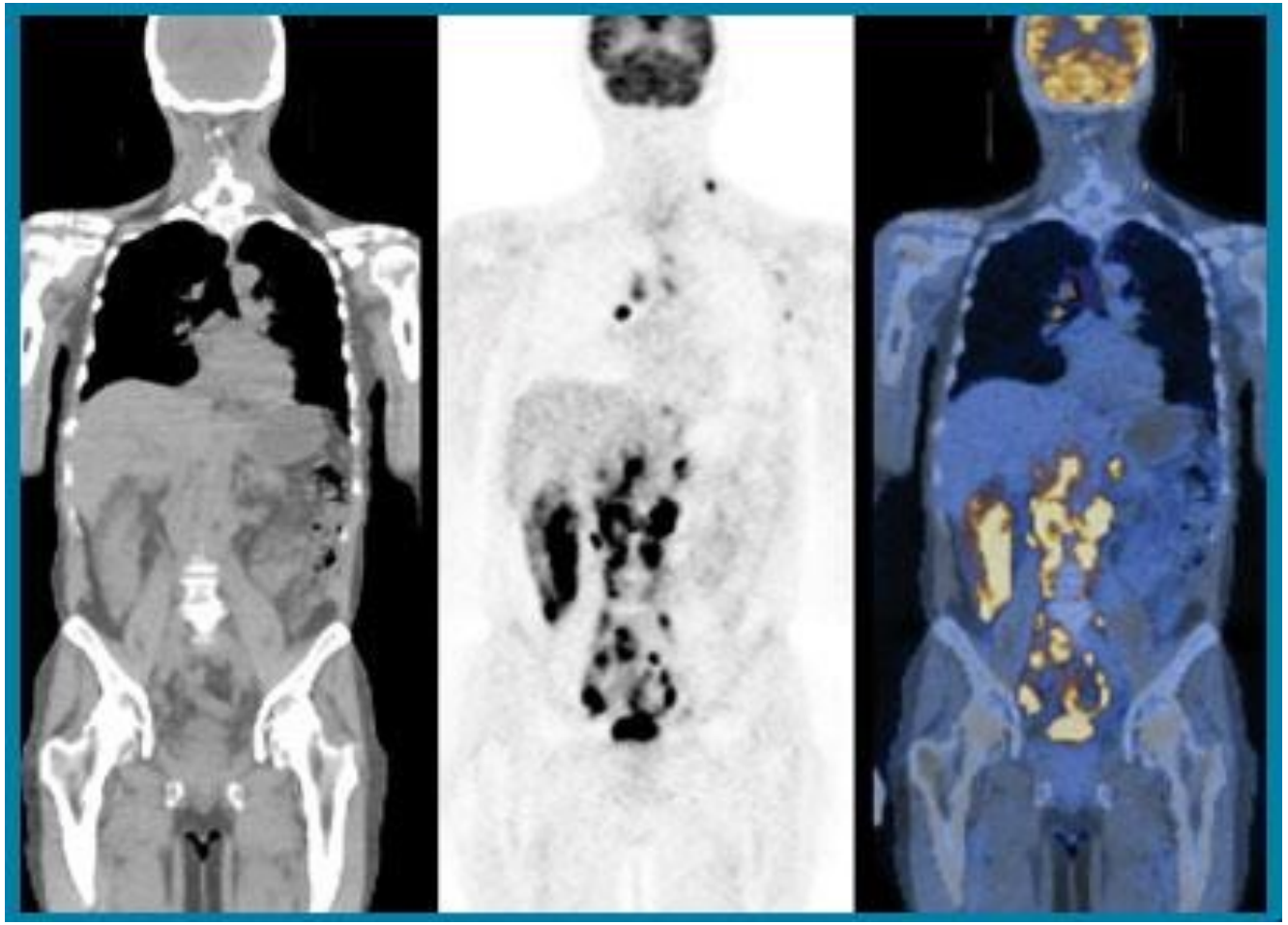
Source: https://openmedscience.com/explorer-pet-ct-scanner-a-total-body-experience/

# PET-CT Scan



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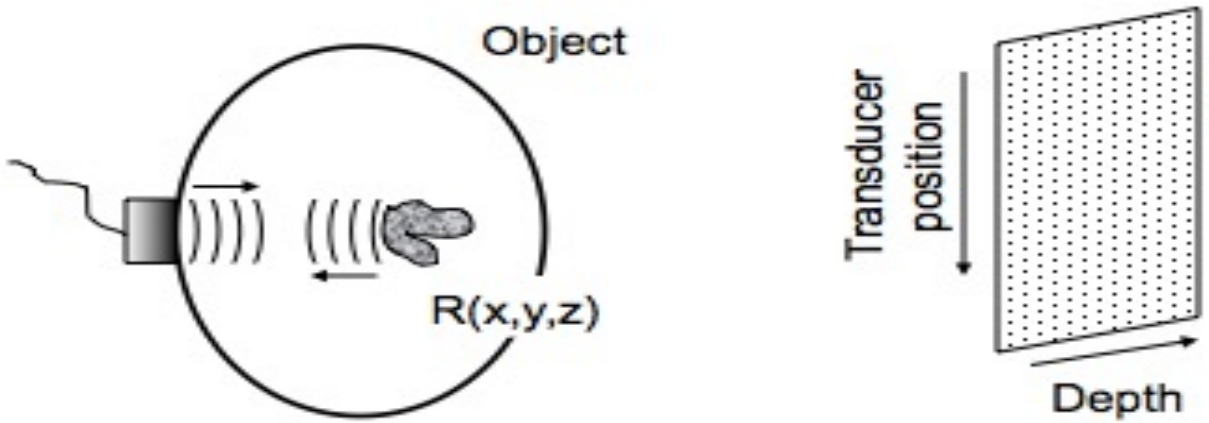
Source: https://busy.org/@massivevibration/technology-never-stands-still-31st-annual-congress-of-the-european-association-of-nuclear-medicine-oct-21-25th-2017-at-the

# Ultrasound Imaging




- Image reflectivity of acoustic wave, R(x,y,z).
- Depth – A function of time (ping-echo)
- Lateral – Focusing of wavefronts
- Direct imaging (e.g. vs. computed) modality – echo data is placed directly into image matrix

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# Ultrasound Imaging

- Issue: Transmit Frequency
  - Increase in frequency reduces wavelength
  - Reduced (improved) resolution size
  - Also improved lateral resolution (diffraction)
  - Increases attenuation
- Issue: Flow
  - Can use Doppler effect to image flow

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# Ultrasound Imaging

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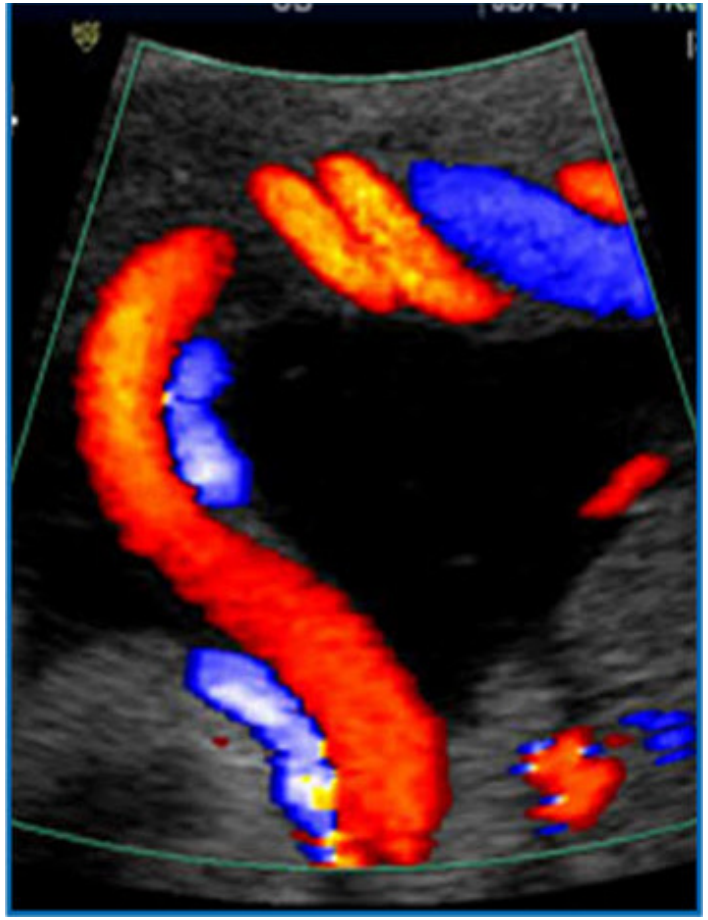
Source: https://marketdeskorg.blogspot.com/2019/02/global-ultrasound-imaging-market.html

# Doppler Ultrasound Imaging



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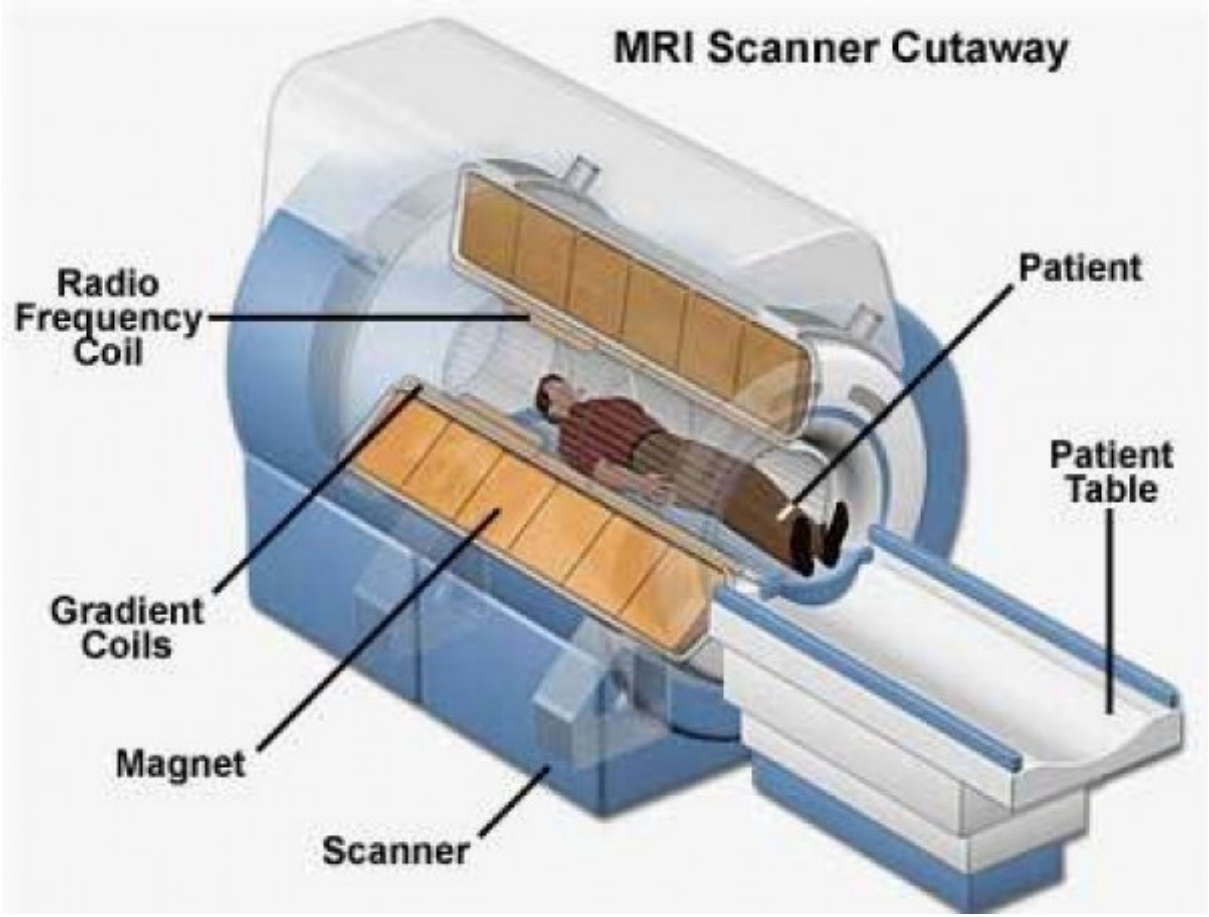
Source: https://fetalmedicine.com/fmf-certification/certificates-of-competence/doppler-ultrasound-1

# Magnetic Resonance Imaging

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Source: https://medium.com/@deepthi.tas/the-story-of-magnetic-resonance-imaging-mri-4ee628d44170

# Magnetic Resonance Imaging

- Atomic nuclei and hydrogen nuclei, $^{1}H$, in particular, have a magnetic moment
  - Moments tend to become aligned to applied field
  - Creates magnetization, m(x,y,z) (a tissue property)
- MRI makes images of m(x,y,z) after switching off the applied magnetic field around the patient.

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# Magnetic Resonance Imaging (cont.)

- The magnetization is excited into an observable state
- Magnetization emits energy at a resonant frequency: (63MHz, 1.5T)
- Frequency is proportional to magnetic field
  - We can create a frequency vs. space variation:

$$\omega(x,y,z) = \lambda\, B(x,y,z)$$

  - Use Fourier analysis to determine spatial location
- Interestingly, $\lambda$ is much larger than resolution – not imaging EM direction, but using its frequency

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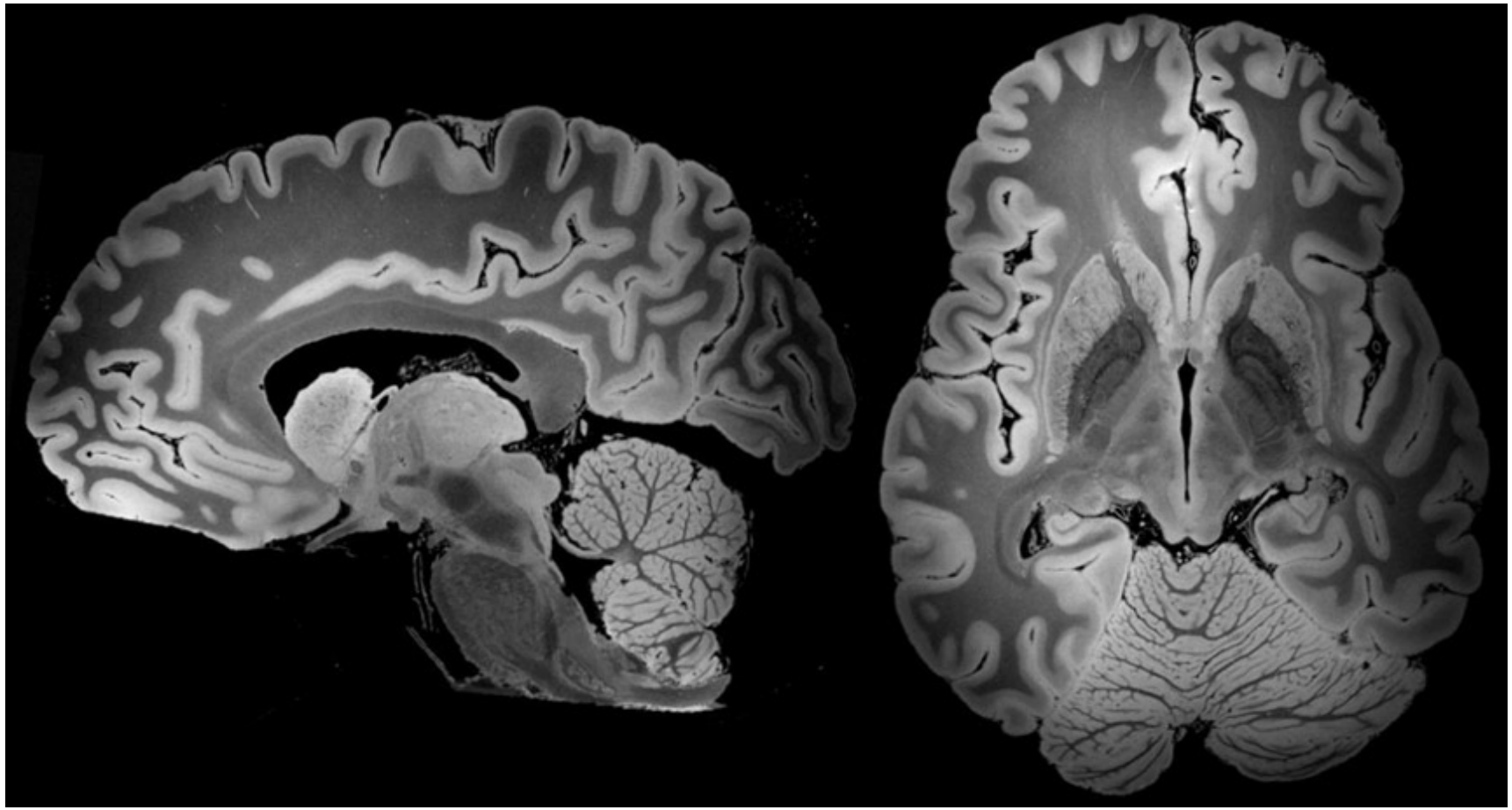
# Magnetic Resonance Imaging



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Source: https://www.sciencenews.org/article/mri-scan-most-detailed-look-yet-whole-human-brain

Thank you for your attention